

Conceptual Model & Relational Schema

ER Diagram

The ER Diagram can be found in a separate file, see [ER_Diagram.pdf](#)

The diagram is drawn in crow's foot notation. Each relationship line is read as:

- || - exactly one
- o{ - zero or many
- ||--o{ - one to many

Entities

Entity	Description
BUILDING	A university building that contains rooms
ROOM	A bookable room within a building
EQUIPMENT_TYPE	A type of equipment that may be present in rooms (e.g. projector)
ROOM_EQUIPMENT	Linking table: records which equipment types a room has, and quantity
ORGANIZATION	A student organization that makes bookings
PERSON	The individual who creates a booking or series
BOOKING_SERIES	The planning intent for a recurring booking (e.g. weekly Wednesday evenings)
BOOKING	One time slot: either a standalone booking or one occurrence of a series

Relationships and Cardinalities

Relationship	Cardinality	Description
BUILDING → ROOM	1 to many	A building contains zero or more rooms; a room belongs to exactly one building
ROOM → ROOM_EQUIPMENT	1 to many	A room has zero or more equipment entries
EQUIPMENT_TYPE → ROOM_EQUIPMENT	1 to many	An equipment type appears in zero or more rooms
PERSON → BOOKING	1 to many	A person creates zero or more bookings
PERSON → BOOKING_SERIES	1 to many	A person creates zero or more booking series
ORGANIZATION → BOOKING	1 to many	An organization is associated with zero or more bookings
ORGANIZATION → BOOKING_SERIES	1 to many	An organization is associated with zero or more series
ROOM → BOOKING	1 to many	A room is the target of zero or more bookings
ROOM → BOOKING_SERIES	1 to many	A room is the target of zero or more series
BOOKING_SERIES → BOOKING	1 to many	A series consists of zero or more individual booking occurrences; a standalone booking has no series

Main Attributes per Entity

1. BUILDING

- **building_id**: surrogate primary key
- **name**: building name
- **address**: physical address

2. ROOM

- **room_id**: surrogate primary key
- **building_id**: FK to BUILDING
- **name or number**: room identifier within the building
- **capacity**: maximum number of people (must be > 0)
- **type**: one of: meeting room, lecture room, studio, workshop room
- **requires_approval**: whether the room requires special access approval (current policy)

3. EQUIPMENT_TYPE

- **Equipment_id**: surrogate primary key
- **Name**: equipment name (unique); e.g. projector, whiteboard, microphone, video camera

4. ROOM_EQUIPMENT (linking table)

- **room_id**: FK to ROOM (part of composite PK)
- **equipment_id**: FK to EQUIPMENT_TYPE (part of composite PK)
- **quantity**: how many units of this equipment type the room has (must be > 0)

5. ORGANIZATION

- **organization_id**: surrogate primary key
- **name**: organization name (unique)

6. PERSON

- **person_id**: surrogate primary key
- **name**: full name
- **email**: contact email (unique)

7. BOOKING_SERIES

- **series_id**: surrogate primary key
- **room_id**: FK to ROOM
- **organization_id**: FK to ORGANIZATION
- **created_by**: FK to PERSON

- **recurrence_rule**: descriptive rule string (e.g. "weekly", "biweekly")
- **day_of_week**: one of: Monday through Sunday
- **start_time**: time of day the slot starts
- **end_time**: time of day the slot ends (must be after start_time)
- **series_start_date**: date of first occurrence
- **series_end_date**: date of last occurrence (must be $\geq$ series_start_date)

8. BOOKING

- **booking_id**: surrogate primary key
- **room_id**: FK to ROOM
- **organization_id**: FK to ORGANIZATION
- **created_by**: FK to PERSON
- **series_id**: FK to BOOKING_SERIES; nullable — NULL for standalone bookings
- **start_datetime**: start of the booking slot
- **end_datetime**: end of the booking slot (must be after start_datetime; same calendar day)
- **status**
- **created_at**: timestamp when the booking was created
- **cancelled_at**: nullable; timestamp of cancellation
- **cancellation_reason**: nullable; short free text
- **requires_approval**: snapshot of the room's approval policy at booking creation time
- **approval_granted**: nullable Boolean: NULL if not required or pending; TRUE if approved, FALSE if rejected

Relational Schema

Tables Overview

Table	Primary Key	Foreign Keys
building	building_id	—
room	room_id	building_id → building
equipment_type	equipment_id	—
room_equipment	(room_id, equipment_id)	room_id → room, equipment_id → equipment_type
organization	organization_id	—
person	person_id	—
booking_series	series_id	room_id → room , organization_id → organization , created_by → person
booking	booking_id	room_id → room , organization_id → organization , created_by → person , series_id → booking_series

Table Definitions

1. building

Column	Type	Nullable	Notes
building_id	SERIAL	NOT NULL	Primary key
name	TEXT	NOT NULL	Building name
address	TEXT	NOT NULL	Physical address

2. room

Column	Type	Nullable	Notes
room_id	SERIAL	NOT NULL	Primary key
building_id	INT	NOT NULL	FK → building
name_or_number	TEXT	NOT NULL	Room identifier
capacity	INT	NOT NULL	Must be > 0
type	TEXT	NOT NULL	
requires_approval	BOOLEAN	NOT NULL	Default FALSE; current room policy

3. equipment_type

Column	Type	Nullable	Notes
equipment_id	SERIAL	NOT NULL	Primary key
name	TEXT	NOT NULL	Unique equipment type name

4. room_equipment

Column	Type	Nullable	Notes
room_id	INT	NOT NULL	FK → room; part of composite PK
equipment_id	INT	NOT NULL	FK → equipment_type; part of composite PK
quantity	INT	NOT NULL	Default 1; must be > 0

5. organization

Column	Type	Nullable	Notes
organization_id	SERIAL	NOT NULL	Primary key
name	TEXT	NOT NULL	Unique organization name

6. person

Column	Type	Nullable	Notes
person_id	SERIAL	NOT NULL	Primary key
name	TEXT	NOT NULL	Full name
email	TEXT	NOT NULL	Unique; only personal field stored

7. booking_series

Column	Type	Nullable	Notes
series_id	SERIAL	NOT NULL	Primary key
room_id	INT	NOT NULL	FK → room
organization_id	INT	NOT NULL	FK → organization
created_by	INT	NOT NULL	FK → person
recurrence_rule	TEXT	NOT NULL	
day_of_week	TEXT	NOT NULL	Monday–Sunday
start_time	TIME	NOT NULL	Slot start time
end_time	TIME	NOT NULL	Slot end time; must be after start_time
series_start_date	DATE	NOT NULL	First occurrence date
series_end_date	DATE	NOT NULL	Last occurrence date; must be ≥ series_start_date

8. booking

Column	Type	Nullable	Notes
booking_id	SERIAL	NOT NULL	Primary key
room_id	INT	NOT NULL	FK → room
organization_id	INT	NOT NULL	FK → organization
created_by	INT	NOT NULL	FK → person
series_id	INT	nullable	FK → booking_series; NULL if standalone
start_datetime	TIMESTAMP	NOT NULL	
end_datetime	TIMESTAMP	NOT NULL	Slot end; must be after start; same calendar day
status	TEXT	NOT NULL	
created_at	TIMESTAMP	NOT NULL	Default NOW()
cancelled_at	TIMESTAMP	nullable	NULL if not cancelled
cancellation_reason	TEXT	nullable	NULL if not cancelled
requires_approval	BOOLEAN	NOT NULL	Snapshot from room at creation time; default FALSE
approval_granted	BOOLEAN	nullable	NULL = not required or pending; TRUE = approved; FALSE = rejected

Check constraint on approval state:

requires_approval	approval_granted	Valid statuses
FALSE	NULL	confirmed, cancelled
TRUE	NULL	pending, cancelled
TRUE	TRUE	confirmed, cancelled
TRUE	FALSE	rejected, cancelled

Note: A booking with **requires_approval = FALSE** cannot sit as **pending**. If no approval is needed, the booking is either confirmed or cancelled directly. This is enforced by the CHECK constraint in the schema (see **schema.sql**).

Linking Tables

1. room_equipment

Resolves the many-to-many relationship between **room** and **equipment_type**. A room can have multiple equipment types; the same equipment type can appear in multiple rooms. The composite primary key (**room_id, equipment_id**) ensures no duplicate entries. The **quantity** column gives this linking table its own structure beyond the relationship itself.

Nullable Columns

Table	Nullable Columns	Reason
Booking	series_id	NULL for standalone bookings
Booking	cancelled_at	NULL if not cancelled
Booking	cancellation_reason	NULL if not cancelled; also optional even when cancelled
Booking	approval_granted	NULL if approval not required or decision not yet made